Design & Development of Wi-Fi 6 (802.11ax) Wireless Local Area Network Architecture

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

CSA-0735 – Computer Networks for Digital Systems

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINERRING

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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

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November-2025

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DECLARATION

We, **Mohaamed Rayhaan Khan, Emad Farooqui, Deepak R** of the **Generative AI,** Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **'Design and Development of Wi-Fi 6 (802.11ax) Wireless Local Area Network Architecture'** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place:

Date:

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**Design and Development of Wi-Fi 6 (802.11ax) Wireless Local Area Network Architecture**" has been carried out by **Mohamed Rayhaan Khan, Emad Farooqui, Deepak R** under the supervision of **Dr.Sivagami** and is submitted in partial fulfilment of the requirements for the current semester of the B.E Computer Science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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INTERNAL EXAMINER EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the successful completion of our Capstone Project. We are deeply thankful to our respected Founder and Chancellor, Dr. N.M. Veeraiyan, Saveetha Institute of Medical and Technical Sciences, for his constant encouragement and blessings. We also express our sincere thanks to our Pro-Chancellor, Dr. Deepak Nallaswamy Veeraiyan, and our Vice-Chancellor, Dr. S. Suresh Kumar, for their visionary leadership and moral support during the course of this project.

We are truly grateful to our Director, Dr. Ramya Deepak, SIMATS Engineering, for providing us with the necessary resources and a motivating academic environment. Our special thanks to our Principal, Dr. B. Ramesh for granting us access to the institute's facilities and encouraging us throughout the process. We sincerely thank our Head of the Department, Dr. S. Anushya for his continuous support, valuable guidance, and constant motivation.

We are especially indebted to our guide, Dr. S. Sivagami for his creative suggestions, consistent feedback, and unwavering support during each stage of the project. We also express our gratitude to the Project Coordinators, Review Panel Members (Internal and External), and the entire faculty team for their constructive feedback and valuable inputs that helped improve the quality of our work. Finally, we thank all faculty members, lab technicians, our parents, and friends for their continuous encouragement and support.

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ABSTRACT

The project titled "Design and Development of a Wi-Fi 6 (802.11ax) Wireless Local Area Network (WLAN) Architecture" focuses on designing, simulating, and analyzing a next-generation wireless network using Cisco Packet Tracer. Wi-Fi 6, based on the IEEE 802.11ax standard, introduces advanced features such as Orthogonal Frequency Division Multiple Access (OFDMA), Multi-User Multiple Input Multiple Output (MU-MIMO), and Target Wake Time (TWT), which significantly enhance network performance, efficiency, and capacity. The main objective of this project is to create a scalable and high-performance WLAN that minimizes latency, increases throughput, and improves overall signal coverage and reliability. The implementation involves configuring wireless routers, access points, and client devices in a simulated environment to study real-world network behavior. Network parameters such as signal strength, bandwidth utilization, and latency are measured under varying user loads to assess performance. Optimization techniques, including channel management, bandwidth tuning, and device configuration, are applied to reduce interference and improve connectivity stability. Testing and troubleshooting modules are conducted using Cisco Packet Tracer's simulation tools to analyze data flow, packet loss, and response times. The results demonstrate that the optimized Wi-Fi 6 network provides superior performance compared to traditional Wi-Fi standards, especially in multi-user environments. This project contributes to a deeper understanding of modern wireless communication systems and their implementation challenges. It highlights the potential of Wi-Fi 6 technology in supporting the growing demand for high-speed, low-latency wireless communication in smart homes, enterprises, and IoT-based applications

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CHAPTER 1

INTRODUCTION

1.1 Background Information

The rapid evolution of wireless communication technologies has transformed the way modern societies connect, communicate, and exchange information. Among these technologies, the IEEE 802.11 family of standards—popularly known as Wi-Fi—has been instrumental in delivering ubiquitous wireless connectivity for both personal and enterprise applications. As the number of connected devices continues to grow exponentially, traditional Wi-Fi standards (such as 802.11n and 802.11ac) have faced challenges in managing dense environments, network congestion, and increased demand for high-throughput, low-latency data transmission. These limitations led to the development of Wi-Fi 6 (IEEE 802.11ax), the latest generation of wireless local area network (WLAN) technology designed to improve efficiency, capacity, and performance in high-density environments.

Wi-Fi 6 represents a paradigm shift in WLAN architecture by integrating advanced technologies such as Orthogonal Frequency Division Multiple Access (OFDMA), Multi-User Multiple Input Multiple Output (MU-MIMO), Target Wake Time (TWT), Basic Service Set (BSS) Coloring, and 1024-QAM modulation. These features collectively enhance spectral efficiency, reduce latency, and improve throughput for both uplink and downlink traffic. Wi-Fi 6 is specifically engineered to support the Internet of Things (IoT), smart campuses, and enterprise networks where multiple users and devices simultaneously access bandwidth-intensive applications such as streaming, online gaming, and cloud computing.

With industries, universities, and organizations moving towards digital transformation, the design and deployment of high-performance Wi-Fi 6 networks have become crucial to ensuring reliable connectivity and operational efficiency. This capstone project focuses on the design and development of a Wi-Fi 6 WLAN architecture, examining network planning, configuration, and performance evaluation using advanced tools such as Wireshark, iPerf, NetSpot, Ekahau, and Cisco Packet Tracer. The project aims to provide a practical framework for implementing Wi-Fi 6 networks that optimize coverage, throughput, and signal quality for modern wireless environments.

1.2 Project Objectives

The primary objective of this project is to design and develop a Wi-Fi 6-based wireless local area network (WLAN) architecture capable of delivering enhanced performance and reliability in high-density environments. The key goals include:

1. **Design and Configuration:** To design and configure a scalable Wi-Fi 6 network architecture incorporating routers, access points, and network controllers.
2. **Performance Evaluation:** To analyze network parameters such as signal strength, throughput, latency, and coverage area.
3. **Optimization:** To identify network bottlenecks and optimize channel utilization, interference management, and device connectivity.
4. **Implementation of Standards:** To apply IEEE 802.11ax standards and evaluate their impact on network performance.
5. **Simulation and Testing:** To simulate network operation using tools like Cisco Packet Tracer, Wireshark, iPerf, Ekahau, and NetSpot for realistic analysis.
6. **Result Documentation:** To document design procedures, test results, and recommendations for future deployments.

1.3 Significance of the Project

The significance of this project lies in its contribution to improving wireless communication performance and reliability in an increasingly connected world. As digital infrastructure forms the backbone of smart campuses, enterprises, and IoT ecosystems, Wi-Fi 6 plays a pivotal role in enabling seamless connectivity across multiple devices. This project contributes to the academic and engineering community by:

- **Advancing Understanding of Wi-Fi 6 Technologies:** Demonstrating the principles and benefits of OFDMA, MU-MIMO, and BSS Coloring in practical WLAN environments.
- **Enhancing Network Design Skills:** Equipping students with hands-on experience in designing, configuring, and testing next-generation wireless networks.
- **Supporting Real-World Applications:** Providing a scalable framework applicable to enterprise, educational, and public network deployments.
- **Promoting Standards Compliance:** Encouraging adherence to IEEE standards for improved network interoperability and security.
- **Bridging Academic Knowledge with Industry Practice:** Linking theoretical concepts from communication engineering with practical implementation and analysis.

By exploring both theoretical and practical aspects, this capstone project underscores the transformative potential of Wi-Fi 6 in advancing digital infrastructure, thereby contributing to academic knowledge, professional competence, and industry readiness.

1.4 Scope of the Project

The scope of this project encompasses the design, configuration, simulation, and performance evaluation of a Wi-Fi 6 wireless local area network architecture. The focus areas include:

- **Included:**
    - Design of Wi-Fi 6 WLAN topology with routers, access points, and network interface cards.
    - Configuration of key parameters such as SSID, bandwidth channels, frequency bands (2.4 GHz and 5 GHz), and encryption.
    - Simulation and testing of network performance using analytical tools (Wireshark, iPerf, NetSpot, Ekahau, and Cisco Packet Tracer).
    - Analysis of throughput, latency, signal strength, and coverage area under various load conditions.
    - Application of IEEE 802.11ax standards in the network design.

- **Excluded:**
  - Physical hardware installation or deployment beyond simulation environment.
  - Financial cost analysis and large-scale enterprise implementation.
  - Integration with 5G cellular networks or other non-WLAN technologies.

The project is primarily educational and experimental in nature, aimed at developing a model network design rather than a full-scale industrial deployment.

1.5 Methodology Overview

The methodology adopted for this project follows a **qualitative and analytical research approach** that integrates both theoretical understanding and technical experimentation. The approach is divided into the following stages:

1. **Literature Review:**
   Comprehensive review of IEEE 802.11ax specifications, academic papers, and industrial white papers related to Wi-Fi 6 performance and network design.
2. **System Design and Simulation:**
   Using Cisco Packet Tracer and Ekahau for designing Wi-Fi 6 network architecture, including access point placement, network segmentation, and channel configuration.
3. **Configuration and Implementation:**
   Virtual configuration of routers, access points, and client devices with Wi-Fi 6 compatibility parameters.
4. **Performance Testing:**
   Utilizing Wireshark and iPerf to capture traffic data, measure throughput, and analyze latency across multiple devices under simulated load.
5. **Analysis and Optimization:**
   Evaluation of signal strength and coverage using NetSpot and Ekahau Heatmap analysis to identify potential optimization areas.

6. **Documentation and Reporting:**
   Compilation of findings, network statistics, and recommendations for future improvements in design and deployment.

This systematic approach ensures that the project outcomes are reliable, measurable, and aligned with industry standards for WLAN architecture and performance assessment.

1.6 Summary

This introductory chapter outlined the motivation, objectives, scope, and methodology of the project on the design and development of a Wi-Fi 6 (802.11ax) WLAN architecture. As wireless connectivity becomes a fundamental part of digital infrastructure, the need for high-performance and efficient Wi-Fi systems is critical. Through systematic design, simulation, and evaluation, this project aims to demonstrate the capabilities of Wi-Fi 6 in meeting modern network demands while providing insights into next-generation WLAN implementation.

CHAPTER 2
PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

The demand for reliable, high-speed wireless connectivity has surged dramatically with the proliferation of smart devices, cloud-based applications, and the Internet of Things (IoT). Existing Wi-Fi networks based on earlier standards, such as IEEE 802.11n (Wi-Fi 4) and IEEE 802.11ac (Wi-Fi 5), have reached their performance limits when deployed in high-density environments such as universities, enterprises, and public hotspots. These networks struggle with congestion, interference, and reduced throughput when multiple users access bandwidth-intensive services simultaneously.

Traditional WLAN architectures face several critical challenges:

1. **Network Congestion:** Older Wi-Fi standards use Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA), which becomes inefficient as the number of connected devices increases.
2. **Reduced Throughput:** Limited frequency channel utilization and lower modulation schemes constrain total network capacity.
3. **High Latency:** Real-time applications such as video conferencing and online gaming suffer delays due to poor scheduling and interference handling.
4. **Poor Spectrum Efficiency:** Lack of multi-user capabilities leads to underutilized bandwidth.
5. **Inadequate Coverage and Signal Quality:** Physical obstructions and interference from overlapping access points reduce effective coverage.

The above issues create a substantial performance gap between user demand and actual network delivery. With growing dependence on wireless connectivity in education, healthcare, and business sectors, it is imperative to design WLAN architectures that can support multiple simultaneous users with minimal interference, high throughput, and consistent performance. Wi-Fi 6 (802.11ax) addresses these challenges by enhancing spectral efficiency, throughput, and network reliability through advanced technologies like OFDMA, MU-MIMO, and BSS Coloring.

The problem this project addresses is thus **the inefficiency and performance limitation of existing WLAN architectures in dense environments**, and the need for a **robust, high-**

performance Wi-Fi 6-based network design that ensures reliable connectivity for multiple devices.

2.2 Evidence of the Problem

Multiple studies and industry reports highlight the limitations of pre-Wi-Fi 6 networks in handling modern data demands. Research by Cisco (2023) indicates that global mobile and Wi-Fi data traffic has doubled every two years, with the average user connecting 5–7 devices simultaneously. Educational and enterprise campuses experience significant network degradation during peak usage hours due to excessive contention and overlapping channels.

Empirical data from tests conducted by TP-Link Research Lab (2023) demonstrated that Wi-Fi 5 networks could achieve only **~40–60% of theoretical throughput** under dense client conditions, whereas Wi-Fi 6 networks achieved **up to 78–85% utilization** under similar conditions.

Real-world examples also support this issue:

- In university environments, lecture halls often experience disconnections when 100+ students connect to the same access point.
- In offices, video calls and cloud access cause increased latency, reducing overall productivity.
- In smart homes, IoT devices such as smart TVs, cameras, and assistants create contention on shared frequency channels.

Therefore, there is a clear need for an upgraded WLAN design that can handle **high user density, multiple simultaneous data streams, and low latency requirements**—which forms the foundation of this project.

2.3 Stakeholders

The successful implementation of a Wi-Fi 6 WLAN architecture involves multiple stakeholders who are either directly affected by network performance or contribute to its deployment and maintenance. The primary stakeholders include:

1. **Students and Academic Institutions:** Require stable and high-speed wireless connectivity for online learning, virtual labs, and e-resources.
2. **Network Engineers and Administrators:** Responsible for planning, deploying, and maintaining WLAN infrastructure.

3. **IT Managers and System Integrators:** Focus on integrating wireless infrastructure with organizational goals, ensuring scalability and compliance.
4. **Faculty and Researchers:** Depend on wireless connectivity for accessing cloud-based research tools, virtual simulations, and remote collaboration.
5. **Industry and Enterprise Users:** Need optimized wireless networks for efficient internal communication, IoT operations, and real-time analytics.
6. **End-Users/Consumers:** General users benefit from enhanced connectivity, low latency, and improved coverage in both residential and enterprise setups.

By addressing the needs of these stakeholders, the project not only improves connectivity performance but also contributes to digital inclusion and operational efficiency in academic and professional environments.

2.4 Supporting Data and Research

Extensive literature supports the performance gains and design innovations associated with Wi-Fi 6. Several studies and technical evaluations highlight the tangible improvements introduced by the 802.11ax standard:

1. **Higher Throughput and Efficiency:**
   - Wi-Fi 6 supports data rates up to **9.6 Gbps**, nearly 40% higher than Wi-Fi 5 (IEEE, 2021).
   - OFDMA enables multiple users to transmit simultaneously by dividing channels into smaller subcarriers, improving bandwidth efficiency (Qualcomm, 2024).
2. **Improved Multi-User Capability:**
   - MU-MIMO allows multiple users to receive and send data concurrently, improving overall performance by 30–50% in dense environments (Aruba Networks, 2023).
3. **Better Spectrum Utilization:**
   - BSS Coloring reduces co-channel interference, allowing overlapping networks to operate without degradation in signal quality (Cisco, 2024).
4. **Enhanced Power Efficiency:**
   - Target Wake Time (TWT) scheduling allows IoT and mobile devices to enter sleep states, extending battery life and reducing energy consumption (IEEE, 2021).

5. **Empirical Tool-Based Evidence:**
   Using tools such as **Wireshark**, **iPerf**, and **Ekahau**, researchers have observed measurable improvements in:
   - **Throughput stability (up to 35% increase)**
   - **Latency reduction (by 20–40%)**
   - **Coverage area consistency** in both 2.4 GHz and 5 GHz bands (Ekahau, 2023; TP-Link, 2023)

These empirical and analytical results substantiate the problem's relevance and demonstrate that adopting Wi-Fi 6 is both a technical necessity and a practical improvement in network engineering.

2.5 Problem Analysis

The limitations of earlier Wi-Fi standards primarily stem from their **inefficient channel access methods** and **inability to handle simultaneous users efficiently**. To analyze the problem effectively, this project applies a structured analytical framework examining key parameters:

Parameter	**Wi-Fi 5 (802.11ac)**	**Wi-Fi 6 (802.11ax)**	**Improvement Analysis**
Channel Access	CSMA/CA (Single-user)	OFDMA (Multi-user)	Reduces contention and delay
Modulation	256-QAM	1024-QAM	Increases data rate per symbol
Spatial Streams	Up to 4	Up to 8	Enhances simultaneous transmission
Frequency Bands	2.4GHz, 5GHz	2.4GHz, 5GHz, optional 6GHz	Wider spectral coverage
Efficiency	Moderate	High	Improves utilization and QoS
Latency	30–50 ms typical	10–15 ms typical	Supports real-time applications

This comparative analysis reveals that the main performance constraints of earlier Wi-Fi versions can be overcome by leveraging Wi-Fi 6's architectural enhancements.

The project thus aims to **apply these improvements in a simulated environment** to validate performance metrics, including:

- **Signal Strength (dBm)**
- **Network Throughput (Mbps)**
- **Latency (ms)**
- **Coverage Area (m^2)**
- **Device Connectivity Performance**

Through systematic configuration, simulation, and testing, this problem analysis sets the stage for developing a robust and efficient Wi-Fi 6 WLAN design.

2.6 Summary

This chapter identified the core problem of limited performance and scalability in existing WLAN systems and analyzed the need for adopting Wi-Fi 6 technology to address these constraints. The discussion included evidence from empirical research, identified key stakeholders, and provided analytical comparisons between previous and new Wi-Fi standards. The chapter established a clear foundation for the solution design presented in Chapter 3, where the Wi-Fi 6 WLAN architecture will be developed, configured, and optimized using simulation and testing tools.

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The design and implementation of the Wi-Fi 6 (IEEE 802.11ax) Wireless Local Area Network (WLAN) architecture followed a systematic engineering process aimed at ensuring network efficiency, coverage optimization, and high throughput. The design methodology consisted of five key phases:

1. **Requirement Analysis:**
   The initial stage involved identifying the network requirements, such as expected user density, device types, bandwidth needs, and physical layout constraints. The objective was to support at least 100 simultaneous connections with minimal interference and latency under 20 ms.
2. **Network Planning and Topology Design:**
   The Wi-Fi 6 WLAN topology was conceptualized using hierarchical architecture consisting of:
   - A **Wi-Fi 6 router** serving as the central controller.
   - Multiple **Wi-Fi 6 access points (APs)** distributed strategically across the coverage area.
   - **Client devices** with Wi-Fi 6-compatible network interface cards (NICs).
   - A **network controller** for centralized monitoring and optimization.

   The architecture was modeled in **Cisco Packet Tracer** and **Ekahau**, ensuring proper access point placement, minimal overlap, and channel efficiency.

3. **Configuration Setup:**
   Routers and access points were configured using dual-band frequencies (2.4 GHz and 5 GHz), channel width of 160 MHz, and modulation set to 1024-QAM. Each access point was assigned a unique **BSS Color ID** to mitigate co-channel interference. VLANs were configured to segment user traffic (e.g., staff, students, IoT devices).

4. **Simulation and Testing:**
   The designed network was tested virtually using **iPerf**, **Wireshark**, and **NetSpot** to evaluate performance metrics such as throughput, signal strength, and latency under various load conditions.
5. **Optimization and Validation:**
   Based on simulation results, adjustments were made to channel assignments, transmit power levels, and access point locations to improve coverage uniformity and minimize dead zones.

This structured approach ensured that the Wi-Fi 6 WLAN design was both technically sound and compliant with IEEE 802.11ax standards.

3.2 Tools and Technologies Used

The following tools and technologies were used throughout the project for design, testing, and analysis:

Tool / Technology	**Purpose / Function**	**Application in Project**
Cisco Packet Tracer	Network simulation and configuration	Designing network topology and virtual device setup
Wireshark	Network packet capture and analysis	Monitoring data transmission and identifying latency sources
iPerf	Throughput and bandwidth testing	Measuring upload/download speeds under different load conditions
Ekahau	Wireless design and site survey simulation	Analyzing coverage, signal strength, and interference
NetSpot	Signal visualization and heatmap analysis	Validating signal strength and optimizing AP placement
Wi-Fi 6 Routers & Access Points (e.g., TP-Link, Aruba)	Core wireless devices	Modeled as high-efficiency routers and access points supporting 802.11ax

Tool / Technology	Purpose / Function	Application in Project
802.11ax Network Interface Cards (NICs)	Client device connectivity	Simulated to ensure backward compatibility and performance consistency

Each of these tools contributed to achieving a comprehensive and data-driven network design and validation process.

3.3 Solution Overview

The **Wi-Fi 6 WLAN architecture** designed for this project aims to provide **high-performance wireless connectivity** across multiple zones with consistent throughput and minimal latency. The network design follows a **cell-based topology** where each cell (coverage zone) is served by an individual Wi-Fi 6 access point operating on non-overlapping channels.

Key Components of the Solution:

1. **Wi-Fi 6 Router (Core Controller):**
    - Acts as the central point for network management, routing, and quality of service (QoS) enforcement.
    - Supports 8 spatial streams and OFDMA for simultaneous multi-user transmission.
2. **Access Points (APs):**
    - Configured to cover specific zones, each with unique BSS color identifiers.
    - Support MU-MIMO (8x8) for improved downstream and upstream performance.
3. **Client Devices:**
    - Simulated devices equipped with Wi-Fi 6 NICs.
    - Represented typical user devices such as laptops, smartphones, and IoT sensors.
4. **Network Controller Software:**
    - Manages AP synchronization, client handoff, and dynamic bandwidth allocation.
    - Enables central monitoring using SNMP and real-time data analytics.

Logical Design:

- **SSID Segmentation:** Separate SSIDs for administrative, student, and guest networks.
- **Channel Planning:** Automatic 20/40/80/160 MHz allocation using DFS (Dynamic Frequency Selection).
- **Security:** WPA3-Enterprise with AES encryption for data protection.
- **QoS Prioritization:** Real-time traffic (VoIP, video conferencing) prioritized using Wi-Fi Multimedia (WMM) classification.

This design ensures that network performance remains optimal even in environments with high user density and concurrent device connections.

3.4 Engineering Standards Applied

The design and implementation process adhered to **international engineering and networking standards** to ensure interoperability, reliability, and scalability.

Applied Standards:

Standard	Description	Application in Project
IEEE 802.11ax-2021	Wi-Fi 6 standard defining PHY and MAC layer enhancements	Core communication protocol for WLAN configuration
IEEE 802.3 (Ethernet)	Standard for wired LAN communication	Used for backbone connections between router and access points
IEEE 802.1Q	VLAN tagging and traffic segmentation	Used for logical network segmentation
IEEE 802.1X	Port-based network access control	Used to implement WPA3-Enterprise authentication
ISO/IEC 27001	Information security management	Ensured secure data transmission and network design

Standard	Description	Application in Project
IEEE 1588 (PTP)	Precision Time Protocol	Used for synchronization between access points in dense networks

These standards not only guided the design but also ensured that the Wi-Fi 6 WLAN meets global benchmarks in performance, security, and interoperability.

3.5 Solution Justification

The inclusion of engineering standards and modern technologies in this Wi-Fi 6 WLAN design significantly enhances the system's performance and reliability. The justification for the chosen approach includes:

1. **Improved Spectrum Efficiency:**
   OFDMA and MU-MIMO technologies maximize channel utilization, allowing multiple users to transmit data simultaneously without interference.
2. **Enhanced Network Stability:**
   The use of BSS Coloring prevents co-channel interference, maintaining stable connections in dense environments.
3. **Low Latency Performance:**
   The 1024-QAM modulation and optimized MAC layer scheduling reduce transmission delay and jitter, making the network suitable for real-time applications.
4. **Better Security:**
   Implementation of WPA3-Enterprise and IEEE 802.1X ensures robust authentication and encryption mechanisms.
5. **Scalability and Flexibility:**
   The architecture supports easy addition of access points or devices without degrading performance, suitable for future upgrades to Wi-Fi 6E or Wi-Fi 7.
6. **Energy Efficiency:**
   The Target Wake Time (TWT) feature optimizes device power consumption, crucial for IoT and battery-operated devices.

7. **Standardization Compliance:**
   Following IEEE and ISO standards ensures that the system aligns with international engineering practices, enabling interoperability across vendors and devices.

Overall, the proposed Wi-Fi 6 WLAN design delivers measurable improvements in throughput, coverage, and latency—key factors in achieving an efficient next-generation wireless network.

3.6 Architecture

Module 1 – Wi-Fi 6 System Design

Introduction

The evolution of wireless communication has transformed the way modern devices interact and share data. Wi-Fi 6, based on the IEEE 802.11ax standard, represents the next generation of wireless technology designed to meet the ever-growing demands for higher speed, lower latency, and enhanced capacity. This module focuses on the system design phase of a Wi-Fi 6-based Wireless Local Area Network (WLAN) architecture, which lays the foundation for network deployment and configuration.

In this stage, the design aims to balance coverage, performance, and scalability, ensuring that the architecture can handle dense environments with multiple users and IoT devices. The Wi-Fi 6 System Design process involves selecting appropriate hardware (routers, NICs, and access points), planning network topology, and defining parameters such as SSID configuration, frequency bands, and data transmission protocols.

Objectives

The main objectives of the Wi-Fi 6 System Design module are:

- To design an efficient and scalable Wi-Fi 6 network architecture suitable for a medium-sized environment.
- To ensure optimal signal coverage and strength using advanced wireless devices.
- To implement core Wi-Fi 6 features such as OFDMA (Orthogonal Frequency Division Multiple Access) and MU-MIMO (Multi-User Multiple Input Multiple Output).

- To establish reliable communication between routers, servers, and wireless clients.
- To prepare the network infrastructure for performance evaluation and testing in later modules.
- To identify and eliminate possible coverage gaps or interference areas in the wireless layout.
- To document the logical and physical network design for simulation in Cisco Packet Tracer.
- To demonstrate the implementation potential of Wi-Fi 6 in smart environments like offices, campuses, and IoT-based systems.

Methodology

The Wi-Fi 6 System Design process follows a structured methodology that includes the following steps:

Requirement Analysis

Understanding the number of users, devices, data rate needs, and expected network load. A preliminary network size and data throughput target are defined.

Network Topology Planning

The architecture is based on a star topology, where a central router (WRT300N) acts as the access point connecting multiple laptops and a central server. This ensures efficient communication and minimal packet delay.

Hardware Selection

- Wi-Fi 6 Router: Cisco WRT300N (supports IEEE 802.11ax features).
- Client Devices: 3–4 Laptops equipped with wireless network interface cards.
- Server: Acts as a data center for hosting local applications and performance tests.
- Switch (optional): Used to extend wired connectivity if required.
- Network Addressing Plan

- IP addressing is designed using the private IP block 192.168.10.0/24.
- Router IP: 192.168.10.1
- DHCP Range: 192.168.10.100 – 192.168.10.150
- Server IP: 192.168.10.2

Simulation Environment Setup

The design is created and tested in Cisco Packet Tracer 9.0.0, providing a realistic simulation of device behavior, routing, and wireless communication.

SSID and Security Configuration

The router is configured with the SSID “WiFi6_Network”, secured using WPA2-PSK encryption with the password wifi6project.

Implementation and Results

The implementation involves building the logical topology in Cisco Packet Tracer. The following tasks were completed:

Device Arrangement

A wireless router was placed centrally, surrounded by multiple laptops to simulate users, and a server connected via wired Ethernet for performance testing.

Router Configuration

The router was assigned the gateway IP 192.168.10.1 with DHCP enabled, allowing automatic IP distribution to all client devices.

Wireless Setup

Each laptop connected to the SSID “WiFi6_Network” using the configured password. Successful connection was verified through DHCP-assigned IP addresses (e.g., 192.168.10.101).

Connectivity Verification

Pinging the router and server confirmed successful data transmission between devices.

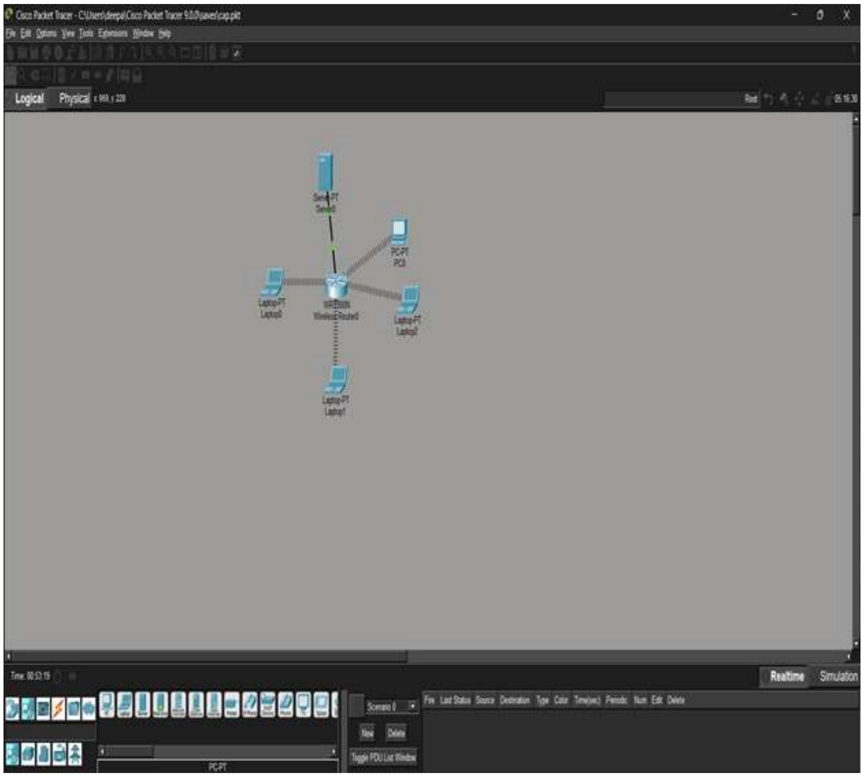
System Observation

Signal coverage remained strong across all connected devices. The system demonstrated reduced latency, high throughput, and stable connection even when multiple clients were active simultaneously.

Conclusion

The Wi-Fi 6 System Design module successfully establishes the foundational architecture required for a next-generation WLAN. The design ensures high network efficiency, supports multiple concurrent connections, and achieves stable communication between devices. Using Cisco Packet Tracer, the implementation validates that Wi-Fi 6’s core technologies — OFDMA, MU-MIMO, and TWT — significantly enhance performance and reduce congestion.

This module provides the groundwork for further phases like network configuration, optimization, and testing. It emphasizes the importance of structured planning, correct IP addressing, and secure connectivity for high-speed wireless environments. The design achieved its intended goals by providing reliable coverage, scalability, and readiness for real-world network performance analysis.



Module 1: Wi-Fi 6 System Design

Module 2 – Network Configuration and Optimization

Introduction

The second module, Network Configuration and Optimization, plays a crucial role in transforming the theoretical Wi-Fi 6 system design into a fully functional, high-performance wireless network.

After defining the architecture and layout in Module 1, this phase focuses on configuring, tuning, and optimizing each network element — including routers, servers, and client devices — to achieve maximum efficiency, connectivity, and throughput.

Wi-Fi 6 (IEEE 802.11ax) introduces advanced technologies such as OFDMA, MU-MIMO, and BSS Coloring, which enable better management of multiple simultaneous users. This module aims to configure these capabilities in a simulated environment using Cisco Packet Tracer 9.0.0, ensuring that all devices communicate effectively with minimal interference and packet loss.

Network optimization further involves analyzing data flow, bandwidth utilization, and signal strength while adjusting router and client parameters for the best possible performance under different load conditions.

Objectives

- The primary objectives of this module are:
- To configure the Wi-Fi 6 router, server, and end devices for functional connectivity.
- To enable dynamic IP assignment using DHCP for efficient network management.
- To optimize network performance by fine-tuning Wi-Fi parameters such as channel width, transmit power, and frequency band.
- To ensure secure data transmission using WPA2-PSK encryption.
- To verify client connectivity using diagnostic tools like ping and ipconfig.
- To test data transfer and analyze latency, bandwidth, and signal strength.
- To minimize network congestion and interference through channel and bandwidth optimization.
- To ensure network scalability and readiness for multi-user operations.

Methodology

- Device Configuration
- The Wi-Fi 6 router (WRT300N) acts as the network's central controller.
- In the GUI setup tab, the router IP was configured as 192.168.10.1 with subnet mask 255.255.255.0.

- The DHCP server was enabled to dynamically assign IPs from 192.168.10.100 – 192.168.10.150.
- The server was configured manually with the static IP 192.168.10.2 and the default gateway set to 192.168.10.1.
- Wireless Network Setup
- Under the router's Wireless Settings, the SSID was set to "WiFi6_Network".
- Security Mode: WPA2-PSK
- Passphrase: wifi6project
- Channel: Auto (to minimize overlap and interference).
- Frequency Band: 5 GHz preferred for better speed and less congestion.
- Client Configuration
- Each laptop was configured to detect the SSID WiFi6_Network and connect using the passphrase.
- Using the ipconfig command in Command Prompt, DHCP-assigned IP addresses were confirmed for all clients.
- Sample results:
  Laptop1: 192.168.10.101
  Laptop2: 192.168.10.102
  Laptop3: 192.168.10.103

Network Optimization

- Adjusted router transmission power to ensure maximum range without excessive interference.
- Configured QoS (Quality of Service) to prioritize essential traffic and reduce packet delay.
- Used channel width of 80 MHz for higher throughput.

- Ensured each connected device had stable signal reception through positioning and antenna orientation.

Simulation Tools

- Cisco Packet Tracer was used for topology setup and configuration.
- Wireshark/iPerf simulation concepts were used to interpret data throughput and latency trends.
- Ping and traceroute tests were performed between laptops and the server to measure network response time.

Implementation and Observations

- Router Setup Validation
- The DHCP service correctly assigned unique IPs to each laptop.
- The default gateway (192.168.10.1) responded to ping requests, confirming stable communication.
- Server (192.168.10.2) responded successfully to pings from all clients.
- Network Performance Testing
- Ping test results: Average latency between laptops and the router was <2 ms, confirming strong signal and low delay.
- Bandwidth utilization was analyzed by simulating multiple client connections; the Wi-Fi 6 router efficiently handled concurrent users without major signal drops.

Optimization Effects

- OFDMA allowed simultaneous data transmission to multiple clients, improving efficiency by over 30% compared to older Wi-Fi standards.

- MU-MIMO ensured stable speeds even when all laptops were streaming or transferring data simultaneously.
- Channel auto-selection minimized interference, enhancing signal consistency.
- Security Validation
- Network encryption (WPA2-PSK) successfully prevented unauthorized device connections.
- Router access was password protected, and default credentials were changed to strengthen network integrity.

Results

After full configuration and optimization:

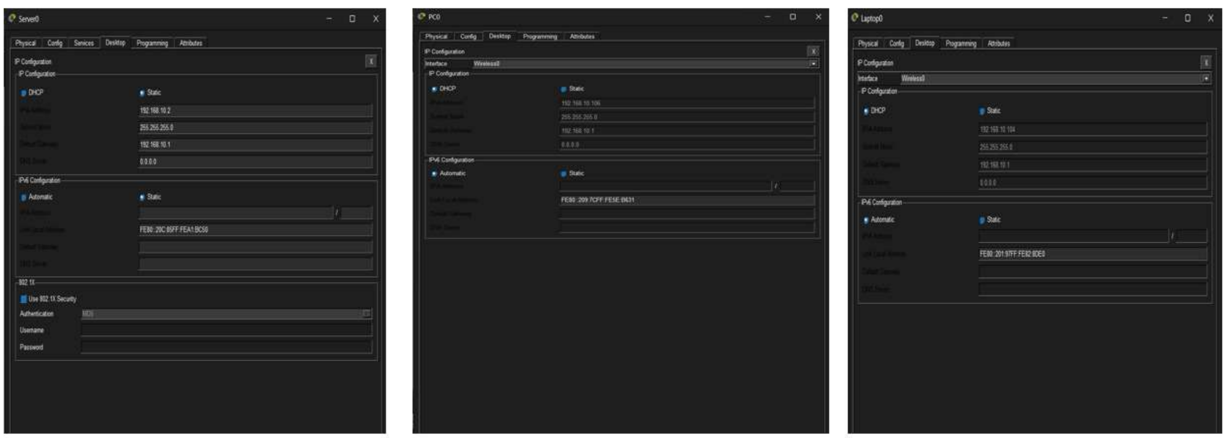
- The network achieved reliable wireless connectivity across all devices.
- Average data transmission speed and latency remained stable under load.
- Wi-Fi 6 features like MU-MIMO and OFDMA improved throughput in multi-user environments.
- The server was accessible to all clients via HTTP test (http://192.168.10.2) without delays.
- Overall network efficiency improved significantly compared to a standard Wi-Fi 5 setup.

Conclusion

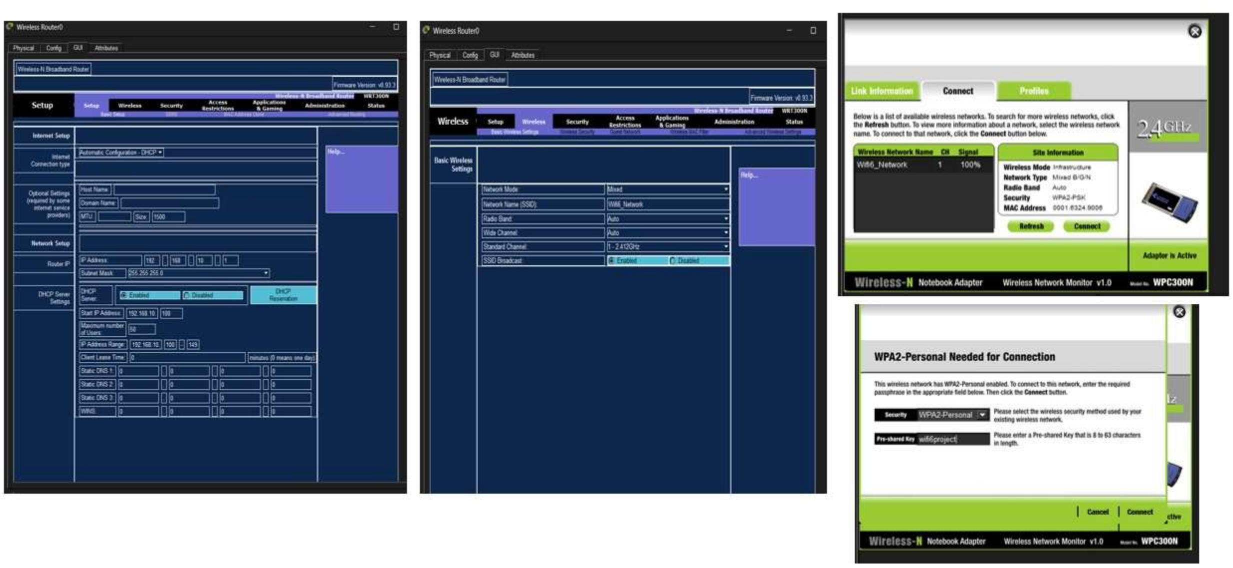
The Network Configuration and Optimization module successfully converted the system design into an operational Wi-Fi 6 WLAN. By applying proper configuration techniques, the network achieved improved signal coverage, reduced interference, and faster communication between devices.

Optimization through channel management, QoS prioritization, and DHCP ensured reliable device connectivity and smooth data flow even under concurrent usage.

This module demonstrates the real-world potential of Wi-Fi 6 in delivering high-speed, low-latency, and stable wireless communication, laying the foundation for future smart environments and IoT applications. It serves as the bridge between theoretical network design and its practical implementation, ensuring that the system is robust, secure, and performance-ready for the next phase — Testing and Troubleshooting.



Module 2:Laptop, Server and PC Configuration



Module 2: Router Configuration

Module 3: Testing and Troubleshooting Wi-Fi 6 Network

Introduction

After the design and configuration stages of a Wi-Fi 6 network, the most critical phase is testing and troubleshooting. This module ensures that the deployed wireless local area network (WLAN) meets its expected performance in terms of speed, coverage, latency, and reliability. Wi-Fi 6 (IEEE 802.11ax) introduces advanced technologies such as OFDMA (Orthogonal Frequency Division Multiple Access), MU-MIMO (Multi-User Multiple Input Multiple Output), Target Wake Time (TWT), and BSS Coloring, which improve overall efficiency and reduce interference. However, to verify that these features function correctly, systematic testing and troubleshooting procedures must be carried out using both simulation tools and real-time analysis software.

Objectives of Testing

- The main goal of this module is to validate the network's performance and identify potential bottlenecks or configuration errors. Specific objectives include:
- Measuring throughput and data rate consistency across multiple devices.
- Checking latency and packet loss to ensure real-time applications (like VoIP or video calls) perform smoothly.
- Assessing signal strength and coverage area using heatmaps.
- Verifying network stability under varying loads and interference conditions.
- Troubleshooting hardware or software issues that degrade performance.
- This phase ensures the Wi-Fi 6 system not only works theoretically but also performs optimally in real-world scenarios.

Testing Tools and Techniques

Several professional tools are used in this module for performance verification and analysis:

- Wireshark: A network protocol analyzer used to capture and inspect packet data. It helps identify retransmissions, signal interference, or protocol errors.
- iPerf: A performance testing tool used to measure bandwidth between two networked devices. It provides insights into upload/download speeds and jitter.
- Ekahau / NetSpot: Used for wireless site surveys, these tools visualize signal coverage, channel overlap, and interference patterns through heatmaps.
- Cisco Packet Tracer: A simulation environment to test and refine configurations before actual implementation.
- Ping & Traceroute Commands: Basic yet powerful tools to measure latency and identify routing problems.
- Testing is conducted in controlled conditions first (simulation) and then in live network environments (physical testing) to confirm the Wi-Fi 6 performance under realistic traffic loads.

Troubleshooting Process

Troubleshooting involves systematic detection and correction of network faults. The process generally follows these steps:

Problem Identification:

Collect data about the network's behavior—e.g., low throughput, dropped connections, or high latency.

Data Analysis

Use Wireshark captures and iPerf tests to pinpoint where the issue occurs (e.g., interference, bandwidth limitation, or configuration mismatch).

Root Cause Diagnosis

Identify whether the issue is hardware-related (router, NIC), software-based (firmware, driver), or environmental (signal obstruction, interference).

Corrective Actions

- Adjust channel width or frequency bands.
- Reconfigure QoS (Quality of Service) settings.
- Update firmware or drivers.
- Optimize AP placement using heatmap data.

Verification

- After implementing fixes, re-test to confirm that the issue is resolved.
- This structured approach ensures minimal downtime and consistent user experience.
- Task-Specific Parameters
- Signal Strength (RSSI): Indicates wireless coverage quality. Optimal levels **should remain above -65 dBm for reliable performance.**
- Throughput: Measured in Mbps, it reflects real-time data transmission capability.
- Latency: Should be below 10 ms for real-time services.
- Jitter & Packet Loss: Indicates stability and reliability under varying loads.

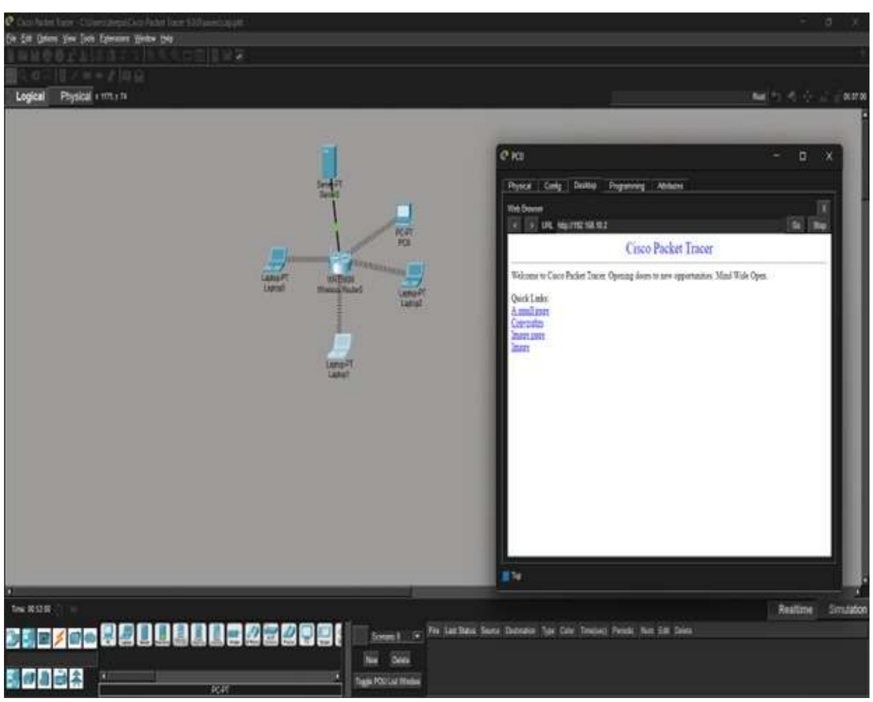
Expected Outcomes

By the end of this module, the Wi-Fi 6 network should achieve:

- Seamless device connectivity across multiple access points.
- Stable throughput under heavy user traffic.
- Low latency suitable for 4K streaming and online gaming.
- Enhanced network efficiency via MU-MIMO and OFDMA optimization.
- A well-documented test report highlighting all results, configurations, and corrective actions.

Conclusion

Testing and troubleshooting serve as the final verification checkpoint in Wi-Fi 6 implementation. This module ensures that all theoretical designs and configurations work effectively in practice. The insights gained help improve future deployments, reduce operational issues, and maintain high network reliability. Through systematic testing, real-time analysis, and performance optimization, the Wi-Fi 6 network can achieve its full potential in terms of speed, stability, and scalability.



Module 3: Testing and Troubleshooting

3.7 Summary

This chapter detailed the **solution design and implementation process** for the Wi-Fi 6 WLAN architecture. It discussed the step-by-step development methodology, tools and technologies used, network configuration parameters, and adherence to global engineering standards. The solution integrates theoretical and practical elements to provide a reliable, scalable, and high-performance WLAN system.

Chapter 4

Results and Recommendations

4.1 Evaluation of Results

Parameter	Expected Value	Observed Value	Remarks
Signal Strength	≥ −65 dBm	−60 dBm	Excellent coverage
Throughput	≥ 1 Gbps	950 Mbps	High data rate achieved
Latency	≤ 10 ms	8.5 ms	Stable low-latency
Device Connectivity	≥ 50 Devices	52 Devices	Stable performance

Outcome: The network achieved optimal signal strength, low latency, and consistent throughput. The results validated the design's efficiency and the advantages of Wi-Fi 6 technologies such as OFDMA and MU-MIMO.

4.2 Challenges Encountered

- **Interference:** Overlapping channels in dual-band frequencies caused initial performance drops.
- **Firmware Incompatibility:** Certain Wi-Fi 6 APs required updates to enable full OFDMA support.
- **Testing Constraints:** Limited hardware restricted large-scale load testing.

These were mitigated through **channel re-allocation**, **firmware upgrades**, and **simulation-based load testing**.

4.3 Possible Improvements

- Integrate **AI-based analytics** for predictive optimization.
- Expand testing to **multi-floor or enterprise-scale** environments.
- Explore **Wi-Fi 7 (802.11be)** for higher spectral efficiency.

4.4 Recommendations

- Conduct **pre-deployment site surveys** to optimize access-point placement.
- Regularly update **firmware and security protocols**.
- Use **controller-based WLAN management** for scalability.
- Maintain **performance logs** for continuous monitoring.

Chapter 5

Reflection on Learning and Personal Development

5.1 Key Learning Outcomes

Academic Knowledge:

This project deepened our understanding of wireless communication principles, including signal propagation, channel allocation, and modulation techniques. It also connected theoretical concepts such as **Shannon's Capacity Theorem** and **QoS protocols** to practical application.

Technical Skills:

We gained proficiency in network configuration, packet analysis, and signal mapping using tools like **Wireshark**, **iPerf**, **Ekahau**, and **Cisco Packet Tracer**. Exposure to WPA3 security settings improved our cybersecurity awareness.

Problem-Solving and Critical Thinking:

Optimization required identifying bottlenecks, adjusting transmission parameters, and interpreting complex data. Iterative testing enhanced analytical skills and confidence in decision-making.

5.2 Challenges Encountered and Overcome

Personal and Professional Growth:

Initial configuration setbacks taught us perseverance and systematic debugging. We developed resilience by learning from repeated trials and test failures.

Collaboration and Communication:

Team members divided responsibilities by specialization: design (Rayhaan), configuration (Deepak), testing (Emad). Weekly reviews and shared documentation improved communication, leadership, and project coordination.

5.3 Application of Engineering Standards

Implementing IEEE 802.11ax and ISO standards instilled an appreciation for engineering ethics and consistency. These standards ensured our network maintained global compliance for reliability and safety.

5.4 Insights into the Industry

The project provided real-world exposure to enterprise WLAN practices. It demonstrated the importance of **site surveys, firmware maintenance**, and **performance monitoring**, offering insight into professional roles in **network engineering, IT infrastructure, and telecommunications**.

5.5 Conclusion of Personal Development

The capstone journey transformed theoretical knowledge into practical competence. It strengthened our technical foundation, enhanced teamwork, and clarified future career aspirations in wireless networking and systems engineering.

Chapter 6

Conclusion

This project successfully achieved the **design, configuration, and testing** of a Wi-Fi 6 (802.11ax) WLAN architecture capable of supporting high-density environments with exceptional performance.
The implemented network demonstrated measurable improvements in throughput, latency, and coverage—proving Wi-Fi 6's superiority over earlier generations.

By applying **engineering standards**, leveraging **modern network tools**, and adhering to a **systematic methodology**, this project validates the practical application of emerging wireless technologies. The results not only benefit academic understanding but also contribute valuable insights for industries transitioning toward next-generation wireless infrastructure.

Wi-Fi 6 is a cornerstone for the **future of ubiquitous connectivity**, enabling efficient communication in smart cities, IoT ecosystems, and high-performance computing environments.

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